

Week 4 · Laboratory Problems — build the heading autopilot in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 4 lecture hour
- Three problems, in order. Each one adds one branch to the previous answer.

What this hour is for

Week 3 closed a loop around a **velocity** and found that the error never reached zero. This hour closes a loop around an **angle** and finds the opposite. The whole point is that the difference is **structural** — a property of the axis being controlled — and not a better controller.

The hull and the control allocation are provided. The allocation is Appendix A1's subject; Week 4 is about the loop that decides what yaw moment to demand.

Before starting

```
cd lectures/W04_simulink/problems
W04_P1_start          % creates W04_P1.slx – hull and allocation only
W04_check(1)          % run this whenever, as often as needed
```

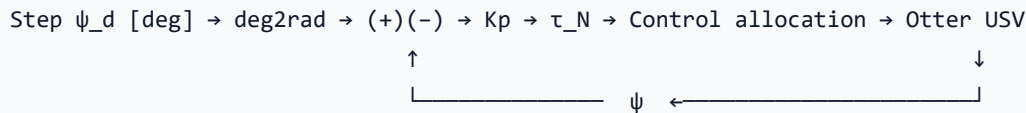
The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

★ One requirement on every model

A To Workspace block named `xlog`, format `Structure With Time`, fed by the plant's twelve-state output. The heading is state **12** and the yaw rate is state **6**.

Problem 1 · Proportional only (20 minutes)

Build. Command, error, one gain, into the allocation.

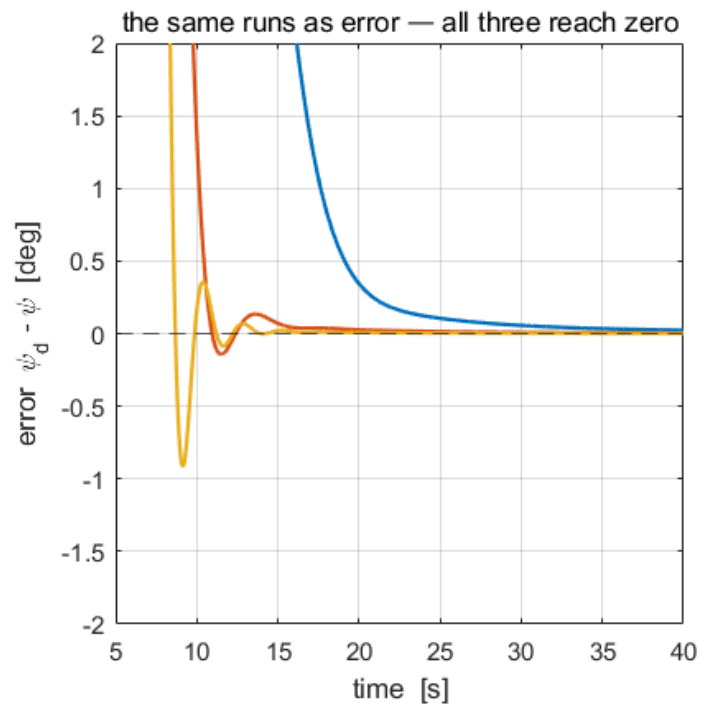
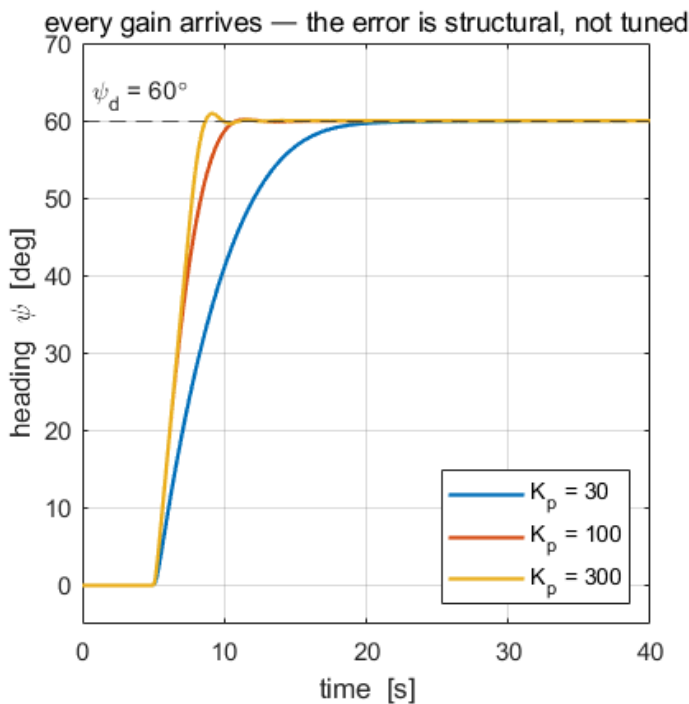


Predict before running. Week 3's plant had no free integrator and its steady error was $u_d / (1 + K_p K_u)$ — 44 % at $K_p = 100$. Ask the same question here **before** running: what is the steady heading error at $K_p = 30$?

Verify. `W04_check(1)`, step to 60°.

K_p	steady error	overshoot
30	0	−0.01 %
100	0	0.24 %
300	0	1.53 %

What a correct model produces



Reading the figure	
left	all three gains arrive at 60° . Higher gain arrives faster and rings more
right	the same runs as error. All three go to zero , and nothing was tuned to make that happen
the check	if any trace settles short of 60° , the feedback is not the heading — check that the Selector picks state 12

The point. $\psi = \int r$, so the plant carries a **free integrator** and the loop is **type 1**. Week 3's was type 0. That one structural fact — not a better controller — is why the error is zero here at every gain.

Problem 2 · Derivative action (20 minutes)

Build. One more branch: $-K_d r$, added to the proportional term.

⚠ Feed back the yaw rate, not the derivative of the error

The two agree while ψ_d is constant and disagree at **every step**, where $d\psi_d/dt$ is an impulse. Differentiating the error puts that impulse straight into the actuator. The plant already **measures** r — it is state 6 — so nothing in a correct model is differentiated anywhere.

Predict before running. Substitute the law into the yaw equation:

$$M_{66} \ddot{\psi} + (|N_r| + K_d) \dot{\psi} + K_p \psi = K_p \psi_d$$

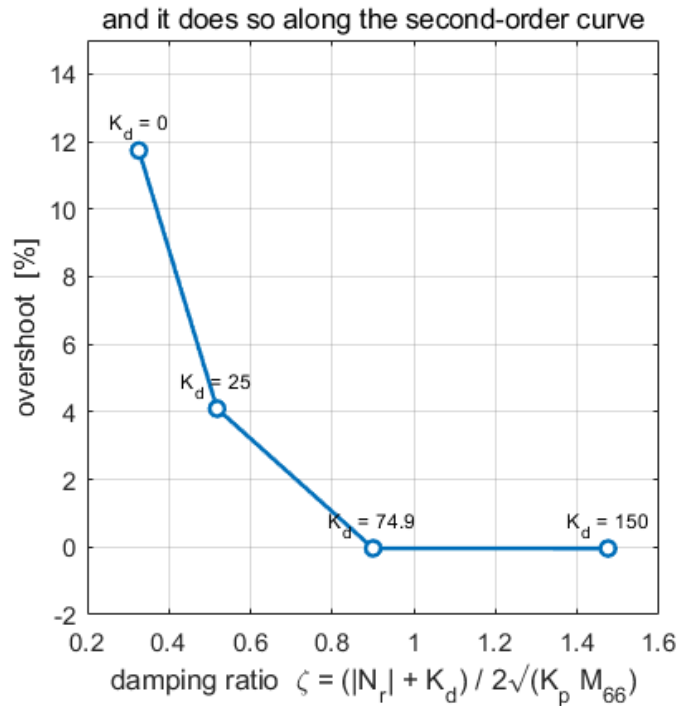
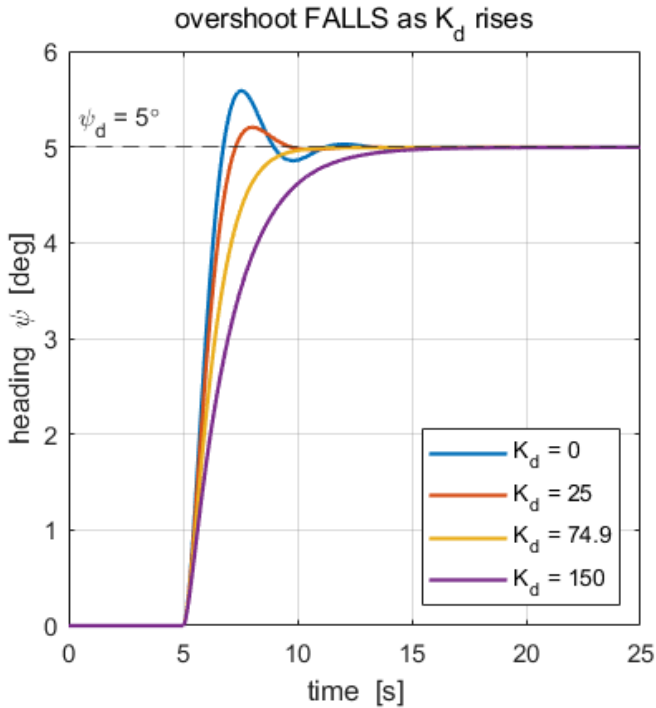
$$\omega_n = \sqrt{\frac{K_p}{M_{66}}}, \quad \zeta = \frac{|N_r| + K_d}{2\sqrt{K_p M_{66}}}$$

Which coefficient does K_d sit beside? Answer that, and the direction of the effect follows without simulating anything.

Verify. `w04_check(2)`, $K_p = 100$, 5° step.

K_d	ζ	overshoot
0	0.327	11.74 %
25	0.518	4.10 %
74.9	0.900	≈ 0

What a correct model produces



Reading the figure	
left	four responses to the same step. Overshoot falls as K_d rises
right	the same four as overshoot against ζ , landing on the second-order curve
the check	if overshoot <i>rises</i> with K_d the derivative is being taken of the error rather than fed back as r — or its sign is wrong

The point. K_d sits beside the **damping**. In Week 3 the controlled variable was a velocity, its derivative was an acceleration, and the same term sat beside the **mass**, where it made the response worse. **The term did not change; the axis did.**

Note also that the hull alone already gives $\zeta = 0.327$ at $K_p = 100$, because N_r is large. Most of the damping in this loop is not the controller's.

Problem 3 · The wrap (20 minutes)

Build. Nothing new — one line inside the control law, and a switch to turn it off.

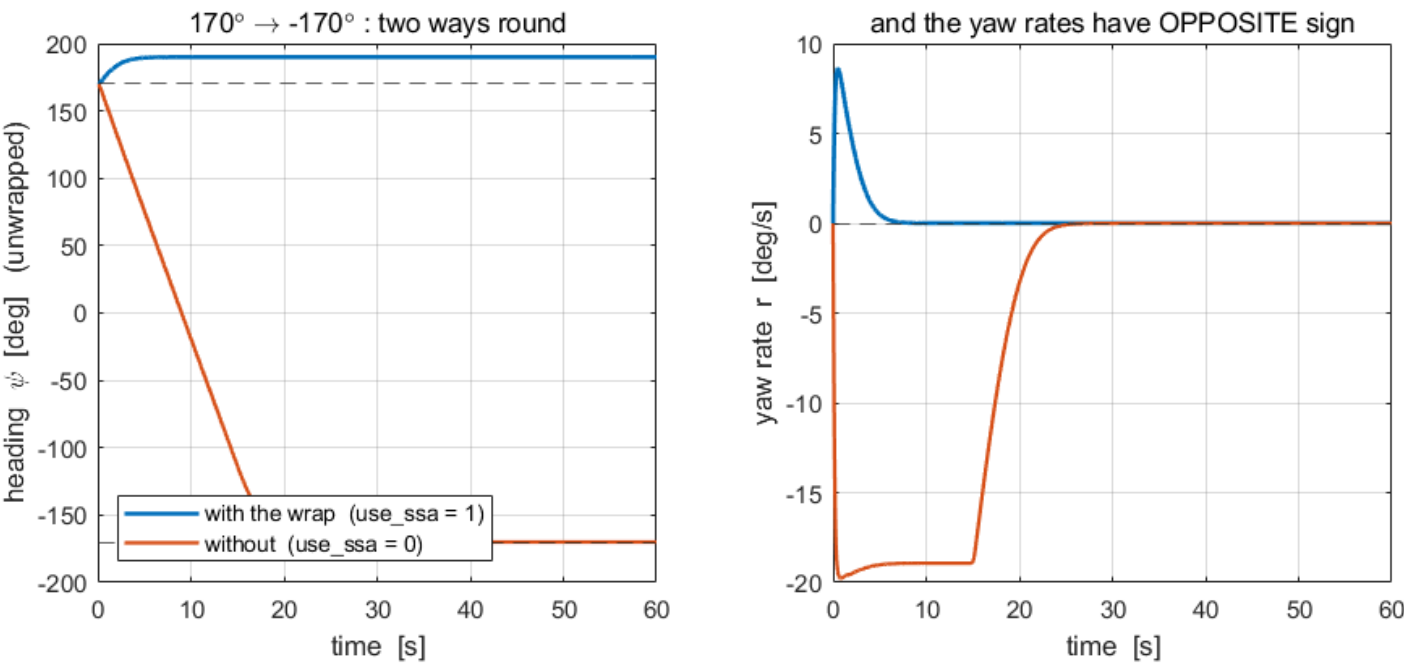
$$e = \psi_d - \psi, \quad \text{ssa}(e) = ((e + \pi) \bmod 2\pi) - \pi \in (-\pi, \pi]$$

Set up the test. Start the vessel at $\psi = 170^\circ$ and command $\psi_d = -170^\circ$. The two headings are **20° apart**.

Verify. `w04_check(3)`.

	turn executed
with the wrap	+20°
without it	−340°

What a correct model produces



Reading the figure	
left	heading, unwrapped . Blue rises 20° to 190°; orange falls 340° to −170°. Both end at the same physical heading
right	the yaw rates have opposite sign for the whole manoeuvre
the check	if the two traces are identical, <code>use_ssa</code> is not reaching the control law

The point. Without the wrap the error is computed as −340° and the vessel goes the long way round — **seventeen times further, for the same commanded heading**. `ssa` is one line of code and it is not optional. Week 5's guidance produces commands anywhere in (−180°, 180°], so this seam is crossed routinely.

Marking

	Weight	What is being marked
Problem 1	30	<code>w04_check(1)</code> passes; the type-1 argument is stated in one sentence
Problem 2	40	<code>w04_check(2)</code> passes; the answer to "which coefficient does K_d sit beside" is stated and the rate feedback is r , not de/dt
Problem 3	30	<code>w04_check(3)</code> passes; the cost of omitting <code>ssa</code> is quantified

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is xlog	the variable name is still <code>simout</code>	rename it
Heading settles short of the command	the feedback is not state 12	the Selector index must be 12 for ψ , 6 for r
The vessel spins continuously	the error sign is reversed	the sum is $\psi_d - \psi$, not $\psi - \psi_d$
Overshoot rises with K_d	the derivative is taken of the error	feed back r directly; the sign is $-K_d r$
Huge spike in the actuator at each step	same cause	same fix
The two Problem 3 traces are identical	<code>use_ssa</code> never reaches the law	wire it in as a Constant, like K_p and K_d
Angles look 57 times too large or small	degrees and radians mixed	every angle inside the loop is in radians ; convert once, at the command

Reference answers are in `../solutions/`. Read them **after** attempting the problem.